FINAL CAPSTONE PRESENTATION

AI-Powered Network Intrusion Detection & Alert Dashboard

A machine learning system that detects malicious network traffic in real time and surfaces it through a live security operations dashboard.

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Systems Integrator

Cole Talarek
ML Developer

Frank Belbusti
Research Lead

01 THE PROBLEM

Enterprise Threats, Small-Business Budget

Nova & Co. is a small online retail store, defended by a three-person security team on a small-business budget.

01

A target like anyone else

Small online stores are scanned and hit daily: DDoS floods, port scans, brute-force logins, and web exploits all target the storefront.

02

Enterprise tools priced out

Commercial IDS platforms and managed SOC contracts cost more than the company's entire annual IT budget.

03

No around-the-clock watch

Three people can't watch traffic 24/7 or triage every alert by hand, so real threats slip by.

04

What Nova & Co. needs

A low-cost, reliable IDS the team can run itself, accurate enough to trust and simple enough to deploy.

NOVA & CO.
ONLINE STORE

02 PROJECT OVERVIEW

What We Built

An end-to-end pipeline that classifies network flows with a trained ML model and streams the results into an interactive dashboard for analysts, with a live capture feed driving the real-time threat map.

DATASET

CICIDS2017

- 2.8M labeled network flows
- 78 features per connection
- Real benign + attack traffic

MODEL

Random Forest

- Selected from 3 candidates
- F1 score of 0.99
- Exported with Joblib

PIPELINE

Detection

- Random Forest classifier
- Predictions + confidence
- Live Scapy capture

OUTPUT

Live Dashboard

- Streamlit + Plotly UI
- Threat map & alert log
- Real-time classification

DETECTING FIVE TRAFFIC CLASSES

BENIGN | DDoS | Port Scan | Brute Force | Web Attack

03 TEAM & ROLES

Who Built What

ML DEVELOPER

Cole Talarek

- Cleaned and preprocessed CICIDS2017
- Prepared the features and encoded labels
- Trained and compared three models
- Exported the selected model via Joblib

SYSTEMS INTEGRATOR

Brian Styrcula

- Designed the system architecture
- Built the live detection pipeline
- Built the Streamlit dashboard
- Integrated all components end to end

RESEARCH LEAD

Frank Belbusti

- Sourced and documented the dataset
- Defined the attack threat scenarios
- Ran the live attack simulations
- Authored documentation and write-up

04 SYSTEM ARCHITECTURE

How Data Flows Through The System

01

Traffic Capture

Scapy captures live packets; CICIDS2017 supplies labeled flows

02

Preprocessing

Flows cleaned to match the training format

03

ML Model

Random Forest scores the CICIDS2017 flows by attack type

04

Detection Engine

Outputs class, confidence, and timestamp per flow

05

Dashboard

Live alerts, threat map, charts, and alert log

HOW THE SYSTEM FITS TOGETHER

The architecture connects every stage from capture to display around the trained model. The Random Forest is validated on the CICIDS2017 benchmark: each flow is preprocessed exactly as in training, run through the exported model, and written out as class, confidence, and severity for the dashboard. A separate live Scapy capture uses lightweight heuristics to drive the real-time threat map. Defining the model-to-dashboard contract was the core integration challenge for the team.

05 DATASET & PREPROCESSING

Turning Raw Flows Into Training Data

THE DATASET

CICIDS2017

Canadian Institute for Cybersecurity benchmark of labeled benign and attack network flows captured over five days.

2.8M
Network flows

78
Features per flow

5
Traffic classes

5 days
Of real capture

PREPROCESSING PIPELINE

1. **Merge & label**
Combined the daily capture files into one frame and normalized the class labels into five categories.

2. **Clean**
Dropped duplicate rows, replaced infinite values, and removed records with missing fields.

3. **Encode labels**
Encoded the class labels into numeric form so the models could train on them.

4. **Train & test split**
Split the data into training and test sets using an 80/20 split.

06 MODEL COMPARISON

Three Models, One Clear Winner

Three classifiers were trained and benchmarked on identical train and test splits, and the top performer was integrated into the live pipeline.

PERFORMANCE COMPARISON (%)

Random Forest

SELECTED

Top-tier accuracy and recall across every attack class, with stable, explainable behavior and the simplest path to deployment.

F1 **99.9856%**

XGBoost

RUNNER-UP

Edged out the top score by a fraction, but added tuning complexity for no practical gain in deployment.

F1 **99.9945%**

Logistic Regression

BASELINE

Fast and simple, but trailed the ensemble models on the harder minority-class attacks.

F1 **99.2379%**

Verdict: XGBoost posted the single highest score (99.9945% F1), but Random Forest matched it to within a hundredth of a percent while being simpler to deploy and already wired into the pipeline. Random Forest was selected for production.

07 RESULTS

Final Model Performance

99.99%
OVERALL ACCURACY

0.996
MACRO F1 SCORE

0.9999
WEIGHTED RECALL

0.01%
OVERALL ERROR RATE

PERFORMANCE BY ATTACK CLASS

Attack Class	Precision	Recall	F1	Support
BENIGN	1.00	1.00	1.00	156k
DDoS	1.00	1.00	1.00	25.6k
Port Scan	1.00	1.00	1.00	18.1k
Brute Force	1.00	1.00	1.00	1.8k
Web Attack	1.00	0.96	0.98	428

WHAT THE NUMBERS SHOW

Near-perfect on volume attacks
DDoS and Port Scan are caught with essentially zero misses.

Strong on the rare classes
Brute Force and Web Attack still score well despite far fewer samples.

Low false alarms
Benign traffic was essentially never flagged as an attack, keeping the alert queue clean.

Validated on held-out data
All metrics come from a held-out test split the model never saw.

08 LIVE DASHBOARD

The Analyst View

AI Threat Detection Dashboard

LIVE

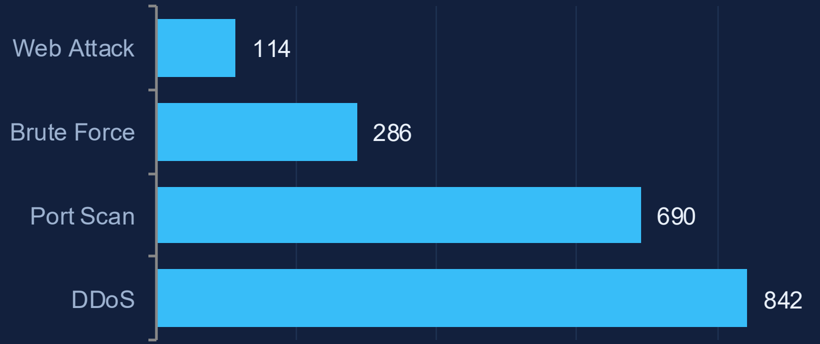
12,480
FLOWS ANALYZED

1,932
THREATS DETECTED

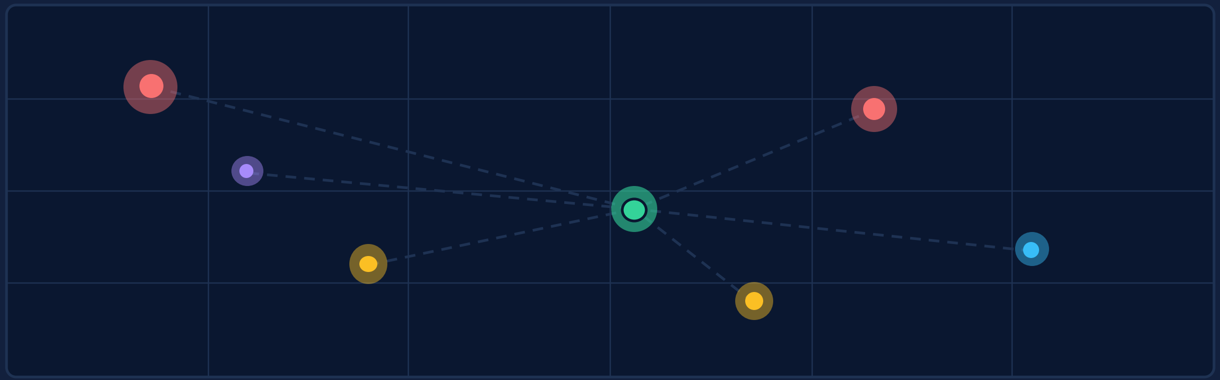
10,548
BENIGN TRAFFIC

96%
AVG CONFIDENCE

THREATS BY TYPE

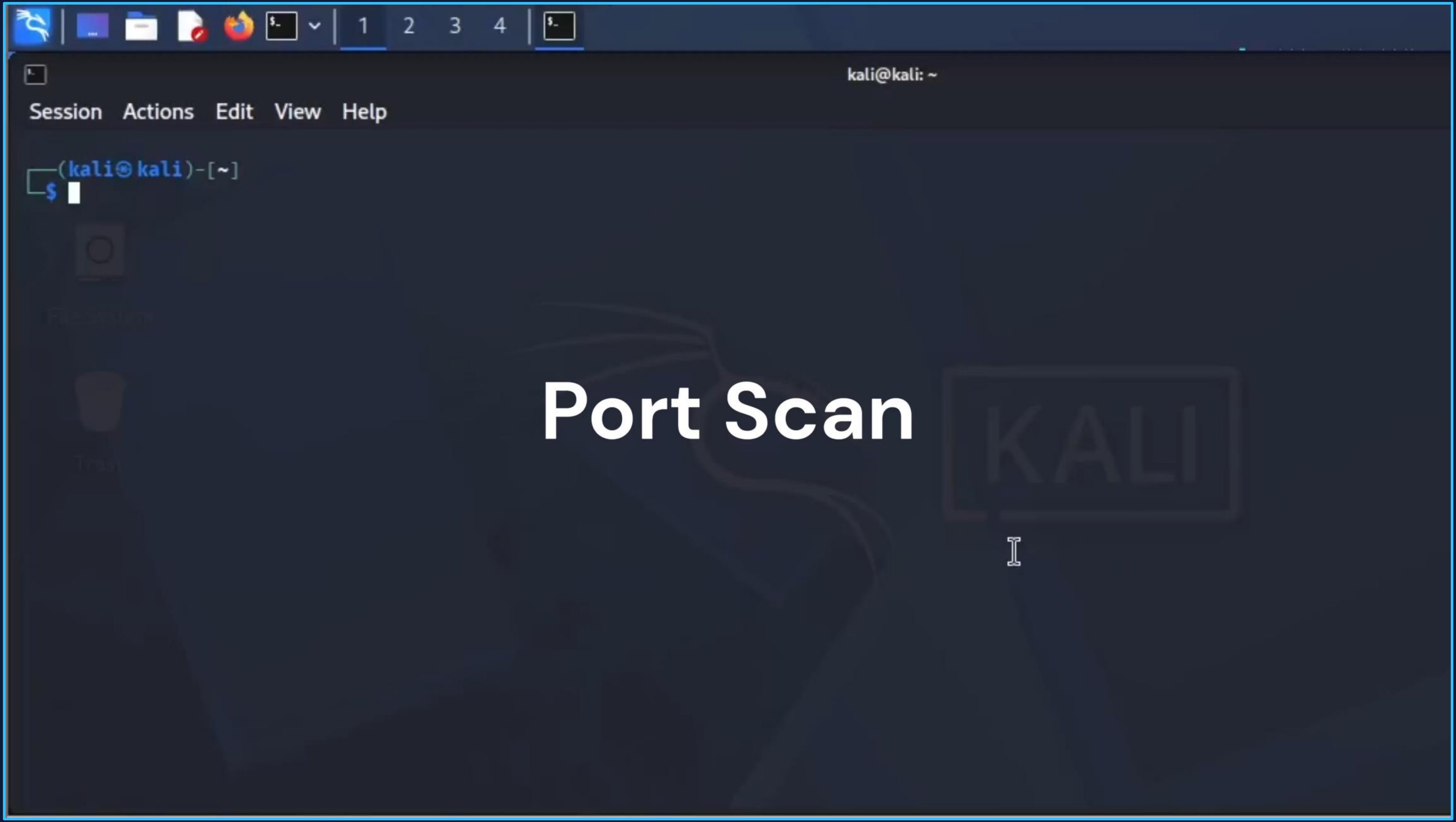


LIVE THREAT MAP · ATTACK SOURCE GEOGRAPHY



Mockup of the dashboard as deployed for Nova & Co. The prediction pipeline exports dashboard-ready records to prediction_output.csv (timestamp, attack type, confidence, and severity), and the KPIs, threats-by-type breakdown, and live source-geography map all update in real time.

LIVE DEMO


1 2 3 4
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$
Port Scan
KALI

EXECUTIVE REPORT

A Report Anyone Can Read

The dashboard is for analysts. The executive report is the plain-language version for everyone else, generated straight from the captured traffic.

Load the alert log

Open the report and point it to the alert log JSON file.

It compiles itself

An executive summary, key metrics, a severity mix, and an attack timeline, all built from the live data.

Built to share

Print or save straight to PDF for leadership. No security background needed to read it.

EXECUTIVE REPORT

AI-Powered Intrusion Detection

120
Total Alerts

67
High Severity

94%
Avg Confidence

EXECUTIVE SUMMARY

During the monitored window the system recorded 29 attack alerts across 4 attack categories. The most frequent attack type was DDoS with 16 events. High-severity activity totalled 22 events from DDoS and Web Attack. Average model confidence across all detections was 97.7%.

09 PROJECT TIMELINE

Nine Weeks, Start To Finish

Wk 1
Setup
Repo, roles, dataset chosen

Wk 2
Data Prep
Cleaned & preprocessed

Wk 3
Modeling
Features & baseline model

Wk 4
Tuning
Compared 3 models, RF chosen

Wk 5
Pipeline
Detection engine built

Wk 6
Dashboard
Full integration

Wk 7
Testing
End-to-end validation

Wk 8
Simulation
Live attacks run

Wk 9
Present
Demo & final report

ALL MILESTONES COMPLETE

What Integration Actually Taught Us

The interface is the hard part

Training a model is bounded; wiring it into a live system is where the real work hides. Most of the effort went to the contract between pipeline output and dashboard input, not the model itself.

Preprocessing must match exactly

Inference data has to be cleaned and structured identically to training data. A single mismatched or reordered column quietly wrecks predictions, so the preprocessing was kept consistent across training and inference.

Metrics can hide weak spots

A 99% overall score looked perfect until the per-class view exposed weaker recall on rare attacks. Slicing by class changed how we read every result after that.

Clear contracts beat heroics

Agreeing early as a team on exact column names, labels, and confidence formats turned final integration into a one-function swap instead of a frantic merge.

11 NEXT STEPS

Where This Goes From Here

The system is complete and demo-ready. These are the directions we would take it toward a production rollout at Nova & Co.

01

Scale the capture layer

Move from single-host Scapy capture to a streaming ingest that can handle production traffic volumes without dropping flows.

02

Expand the threat coverage

Add infiltration, botnet, and zero-day-style classes, and retrain on newer datasets to keep pace with evolving attacks.

03

Wire up real alerting

Push high-confidence detections into email, Slack, or a SIEM so analysts are notified without watching the dashboard.

04

Containerize and deploy

Package the pipeline and dashboard in Docker for repeatable deployment, with the model served behind a versioned endpoint.

PROJECT COMPLETE

Model trained. Pipeline built. Dashboard live.

For Nova & Co., we trained and compared three models, integrated the strongest into a live detection pipeline, and built the analyst dashboard that streams classified threats in real time.

0.99
FINAL F1 SCORE

5
THREAT CLASSES

Real time
CLASSIFICATION

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Thank you. Questions welcome.